



Model-Based Systems Engineering Approach for Simulating Urban Air Mobility Maturity Level 5 (UML-5) Operations

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A mature state of Urban Air Mobility (UAM) will see tens of thousands of simultaneous, heterogeneous passenger-carrying and cargo-carrying operations in urban areas. To describe the evolution of UAM state of operations, NASA has outlined 6 UAM Maturity Levels (UML1 - UML6). The FAA and NASA have each released Concept of Operations for UML-1 and UML-4, respectively. They outline the UAM operational framework, the role of stakeholders, infrastructures, and the level of automation. A more mature states of operations begin at UML-5, where autonomous networks are highly integrated into ground-based remote piloting operations. Safe and Secured Assured Autonomy (S2A2) is a NASA University Leadership Initiative (ULI) project aimed at developing autonomy-enabled algorithms for perception, sensing, coordination, planning, navigation, detecting and responding to malicious attacks on the cyber-physical systems. Verification and validation processes are being developed to certify the algorithms. The challenge remains to architect a systems integration pipeline to fit all the different aspects above into a UAM system-of-systems context. This paper uses Model-Based Systems Engineering (MBSE) approach for a UML-5 operational environment where S2A2 algorithms will cohesively work together while performing their independent functions. We use SysML to define the UAM operating context, its stakeholders, their state transitions, and functions for a nominal UAM operations scenario. These are presented via Block Definition Diagrams (BDDs), Internal Block Definition Diagrams (IBDs), State Machine Diagrams (STMs), and Activity Diagrams (ADs).

Nomenclature

AD	=	Activity Diagram
BDD	=	Block Definition Diagram
DEP	=	Distributed Electric Propulsion
eVTOL	=	Electric Vertical Takeoff and Landing
FO	=	Fleet Operator
IBD	=	Internal Block Diagram
MBSE	=	Model Based Systems Engineering
NAS	=	National Airspace System
OC	=	Onboard Controller
PIC	=	Pilot-In-Command
PSU	=	Provide of Service for UAM
ROPE	=	Resources, Operations, Policies, Economics
S2A2	=	Secure and Safe Assured Autonomy
STM	=	State Machine Diagram
SVO	=	Simplified Vehicle Operations
TC	=	Technical Challenge
UAM	=	Urban Air Mobility
UML	=	UAM Maturity Level
V2V	=	Vehicle-to-Vehicle

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I. Introduction

ADVANCEMENTS in Distributed Electric Propulsion (DEP) and autonomy are enabling the future of electric vertical takeoff and landing (eVTOL) vehicles to thrive. The mature state of Urban Air Mobility (UAM) will see tens of thousands (10,000s) of simultaneous passenger-carrying and cargo-carrying, heterogeneous UAM operations in urban areas. To chart the path towards this vision, NASA has introduced UAM Maturity Levels 1 - 6 (UMLs) to define the state of the UAM environment at different phases of integration into the the National Airspace System (NAS).

UML-1 will involve late-stage certification testing of new eVTOL vehicles and operational demonstrations in limited environments. The FAA has released Concept of Operations for UML-1 [1] which outlines the operational framework, the stakeholders, and the infrastructure necessary for UAM operations. This phase will have a Pilot-in-Command (PIC) on-board manually flying the aircraft. Furthermore, NASA has released Concept of Operations for UML-4 [2] which outlines medium density and complex operations in urban areas with responsible automated systems. This phase will see 100s of simultaneous operations in urban centers. UML-4 will enable Simplified Vehicle Operations (SVO) [3, 4]. Because autonomy is integrated into the operations, the amount of training required for pilots and the cost of hiring pilots is reduced [5]. The Concept of Operations for UML-4 also details the role of stakeholders in the UAM ecosystem.

A mature state of UAM operations is planned to occur in UML-5 and UML-6. UML-5 will see high density and complex operations with highly-integrated automated networks [6]. In this level, UAM aerodrome networks are highly connected. On-board pilots and SVO will be replaced with ground-based remote pilots, with M:N allocation ratio (M aircraft to N remote-pilots). UML-6 will see ubiquitous operations with system-wide automated optimization. There will be 10,000s of simultaneous operations where the only capacity limitations are due to the number of aerodromes. There will be neither on-board pilots nor actively operating remote-pilots. Instead, they will be replaced with supervisory pilots who only monitor operations from ground.

The challenges of realizing the goals of the mature state of UAM operations (i.e., UML-5 and UML-6) lie in developing safe and secure assured autonomy algorithms [7]. Safe and Secure Assured Autonomy (S2A2) is a NASA University Leadership Initiative (ULI) funded project with collaboration from various research institutions and industry partners focused on developing and integrating autonomy-assured algorithms into the UAM environment [8, 9]. The aspects of safety incorporate autonomy-enabled sensing, perception, coordination, planning, and navigation algorithms [10, 11]. The aspects of security incorporate secure and resilient cyber-physical systems that detect malicious attacks on the aircraft [12]. Furthermore, verification and validation processes are set in place to certify the developed algorithms before they are integrated into the vehicle [13]. In addition to verification and validation, these algorithms will have fit into the whole UAM system-of-system operating environment.

The Concept of Operations reports for UML-5 and UML-6 are yet to be developed. This paper begins defining the aspects of safe and secure assured autonomy, its system integration, for UML-5. The goal is to use Model-based Systems Engineering (MBSE) approach for UML-5 operational environment that defines UAM operating context, its stakeholders, their states, and actions. The scope of this paper covers only those which fall under NASA S2A2 project scope. We aid the integration of the developed algorithms for S2A2 within this UAM system context. We study how the outcome of these algorithms trigger the interactions between UAM stakeholder. Furthermore, we will identify the impact of UAM system-of-system performance from the autonomy-assured algorithm requirements.

Due to the complex problem space of scaled UAM operations, the UAM system-of-systems lexicon must be defined to ensure consistent and clear communication between researchers and industry partners. A Resources, Operations, Policies, Economics (ROPE) table is created to ensure consistent taxonomy. Existing literature of UML-1 and UML-4 Concept of Operations are utilized to map out a path towards UML-5. The ROPE table developed by Purdue researchers in identifying operations limits for passenger carrying UAM operations [14] is used as a reference. The operational environment and performance targets for safe and secure assured autonomy algorithms are identified. To achieve all of the above, we use SysML on *Dassault Systemes MagicDraw* to create MBSE diagrams specific for the use cases of UML-5 operations.

II. Literature Review

A. Urban Air Mobility

UAM is an ambitious concept of utilizing advanced air transportation to carry passengers and cargo over short distances in densely populated cities. UAM is a subset of the Advanced Air Mobility (AAM), a NASA-FAA-Industry collaborative initiative that aspires to use innovative air vehicles for passenger and cargo transportation spanning various ranges (local, regional, intra-regional, etc.). Due to the growing urban population and traffic congestion in the streets,

the UAM concept has gained immense traction, received attention and support from industry and government bodies, alike. However, to bring this exciting prospect to fruition, lots of research and development is required about safety, security, operational framework, and guidelines for various stakeholders involved.

The NASA “UAM Vision Concept of Operations (ConOps) UML-4, Version 1.0” [2] provides an outline of the UAM concept, the various maturity levels, the operating environment, the organizational framework, and the challenges to be overcome for realizing this concept. At full maturity, the UAM framework will allow the urban population to benefit from autonomous and semi-autonomous air vehicles for commute, package delivery, etc. The eVTOL market is emerging as the best suited due to its low operational costs compared to traditional helicopter [15], [16]. NASA UML’s are summarized as follows:

- 1) **UML-1:** Late-Stage Certification Testing and Operational Demonstration in Limited Environments.
- 2) **UML-2:** Low Density and Complexity Commercial Operations with Assistive Automation.
- 3) **UML-3:** Low Density, Medium Complexity Operations with Comprehensive Safety Assurance Automation.
- 4) **UML-4:** Medium Density and Complexity Operations with Collaborative and Responsible Automated Systems.
- 5) **UML-5:** High Density and Complexity Operations with Highly-Integrated Automated Networks.
- 6) **UML-6:** Ubiquitous UAM Operations with System-Wide Automated Optimization.

The UAM system and operations are composed of the following roles, termed ‘UAM pillars’ by NASA [2].

- 1) **Airspace System Design and Implementation:** This includes the design of the airspace and the UAM Aerodrome, outlining the Communication, Navigation, Surveillance, and Information (CNSI) infrastructure, and specifying the roles and responsibilities of governments, private service providers, and fleet operators.
- 2) **Individual Aircraft Management and Operations:** This encompasses flight management, aircraft operations, certification approval, ground operations, and maintenance.
- 3) **Airspace and Fleet Operations Management:** The major responsibility of this pillar is to ensure safe, efficient, scalable, and resilient flight operations, along with fleet management. Apart from routine activities, the Fleet Operator must be prepared to mitigate off-nominal situations such as inclement weather or aircraft system failure.
- 4) **Aircraft Development and Production:** This pillar is responsible for the design, certification, manufacturing, and supply of aircraft to the UAM system. This pillar’s important factors are aircraft safety, tolerance, efficiency, aircraft noise, passenger cabin safety, comfort, etc.
- 5) **Community Integration:** This pillar handles the challenges of ensuring public acceptance, leveraging the available supporting infrastructure, operational integration among the various sections of the UAM airspace, and mediating between the different pillars. Another important responsibility is to ensure that the UAM systems follow all the government standards and environmental regulations.

Complementary to the NASA Concept of Operations is the “FAA Urban Air Mobility (UAM) Concept of Operations” [1], developed by the FAA NextGen office. This document details the need for the incorporation of UAM operations into the National Airspace System (NAS), holding it in the context of Air Traffic Management (ATM) and Unmanned Aircraft System (UAS) Traffic Management (UTM). With advancements in obstacle detection-and-avoidance algorithms, cloud networks, mobile phone capabilities, etc., the feasibility and demand for ‘air taxis’ and other UAM-based services have increased drastically. At this point, it is essential to start developing a proper operational framework and infrastructure for UAM. With several air vehicles flying in a relatively small urban air space, traffic management will be of utmost importance in the UAM operation scenario [17]. The UAM operational environment defined by the FAA includes some important terms, which have been defined below.

- **UAM Aerodrome:** A location for the arrival or departure of UAM vehicles.
- **UAM Aircraft:** An aircraft that can execute UAM operations.
- **UAM Corridor:** A three-dimensional airspace volume defining specific routes; based on aircraft category and performance metrics, it may operate within or cross this Corridor; tactical ATC separation services are not provided.
- **UAM Operation:** The transport of people or goods from one aerodrome to another using UAM aircraft.
- **UAM Operator:** The organization or entity responsible for the execution and management of a UAM operation.
- **UAM Operational Intent (flight intent):** Key information of a UAM operation, such as identification, intended UAM Corridor(s), aerodromes, arrival and departure times, etc.

Important roles and their main responsibilities assigned by the FAA are listed below.

- **FAA:** The FAA is the regulatory authority for all aircraft operations, including the UAM category. It develops and modifies regulations to facilitate UAM operations, defines, maintains, and regulates UAM Corridors, and sets the requirements for operation within and across these Corridors. Traffic control guidelines may be suggested by FAA, although the tactical separation within UAM Corridors is allocated to the UAM community and not ATC. FAA provides data sources to UAM operations and can support non-nominal operations.
- **UAM Operator:** The operator is tasked with all aspects of UAM operation executions and regulatory compliance. This involves data sharing for safety and situational awareness (including weather, airspace constraints, advisories, etc.), strategic deconfliction, and support for separation management.
- **Pilot in Command (PIC):** Currently, the PIC is considered to be the person aboard the UAM aircraft, but at higher maturity levels, the aircraft can be remotely piloted or autonomous. The PIC is responsible for the operations and safety during flight.
- **Provider of Services for UAM (PSU):** A PSU is an entity that supports UAM operators to meet UAM operational requirements while maintaining the airspace's safety, security, and efficiency. Its responsibilities include distributing relevant data, analyzing and verifying flight intent, communication via the PSU network, determining UAM Corridor status, etc.
- **PSU Network:** A PSU Network is the collection of connected PSUs that share important information to facilitate communication among federated UAM actors and overall situational awareness.
- **Supplemental Data Service Providers (SDSPs):** SDSPs are used by UAM operators and PSUs to obtain supporting data.
- **UAM Aerodrome:** A UAM aerodrome must meet the capability requirements to support UAM departure and arrival operations. It provides real-time pad availability status and other data for UAM operations.
- **UAS Service Supplier:** These entities support UAS operations through the UTM framework. They interact with PSU when a UAS aircraft needs to cross UAM Corridors and can provide support to UAS and UAM operations during off-nominal conditions.
- **Other NAS Airspace Users:** These include all non-UAM NAS users. They are responsible for being aware of the relevant performance requirements for operating, crossing, or avoiding active UAM Corridors.
- **Public Interest Stakeholders:** These include local law enforcement, US federal agencies, and other entities that have permission from the governing bodies to access UAM operational information.

The FAA ConOps further elaborates the notional architecture, nominal and off-nominal operational guidelines, and UAM implementation strategies.

B. Model Based Systems Engineering

International Council on Systems Engineering (INCOSE) defined Model-Based Systems Engineering (MBSE) [18] as “the formalized application of modeling to support system requirements, design, analysis, verification and validation activities beginning in the conceptual design phase and continuing throughout development and later life cycle phases.” It may be considered a technological evolution of document-based systems engineering. MBSE focuses on models to develop the system design by combining a bottom-up and top-down approach into a single platform. While MBSE had been in use for decades by NASA, Boeing [19], etc., it has recently gained traction in industry and research organizations with an increasing global inclination towards digital modeling environments. MBSE may be implemented using Mechanical and Electrical Computer Aided Design (M-CAD, E-CAD), Systems Modeling Language (SysML), Unified Modeling Language (UML), etc. [20]. In an MBSE-based approach, a system model comprises component/structural, functional/behavioral, performance, and various analysis models. These can collectively help system engineers understand the overall impact of each design element, easily track flaws or errors, represent complex relationships in a simplified manner, and work as a team on various subsystems of a unified system.

One of the enabling technologies for MBSE is SysML. SysML is a general-purpose graphical modeling language for systems engineering applications [21]. It was developed by INCOSE, the Object Management Group (OMG), and the ISO AP 233 workgroup to modify the Unified Modeling Language (UML) for systems engineering applications. SysML can handle complex systems that involve hardware, software, data, personnel, procedures, and facilities [20]. It can simplify the design, analysis, verification, validation, operation, and maintenance of such systems. SysML stands on four ‘pillars’, namely, structure, behavior, requirements, and parameters. The system structure relies on block definition diagrams and internal block diagrams, which specify the system’s structural properties, hierarchy, and components.

The system behavior follows use-case, Activity, Sequence, and State Machine Diagrams, which describe the function, operational states, and interactions of the structural elements. The requirements diagram enables systems engineers to relate the requirements with the system models. Finally, a parametric diagram imposes constraints on the systems via logical and mathematical expressions. SysML also allows the use of allocation relationships to define functions and interconnections. SysML and MBSE have been used extensively in this project to achieve smooth and cohesive UAM Operations.

III. ROPE Table, System-of-System Lexicon, Stakeholders

The trade space of a system-of-systems may consist of the Resources, Operations, Policies, and Economics (ROPE), which can be summarized in a ROPE table [22]. This table aims to avoid lexicon confusion in trans-domain applications and uncover key behaviors and sources of complexity. ROPE table identifies and describes various systems contexts, hierarchies, and interrelationships. Before constructing the ROPE table for the UML-5 operating context, the stakeholders for UML-5 are mapped from the NASA Concept of Operations for UML-4 (Table 1). Since our focus is on safe and secure assured autonomy in UML-5, the number of key stakeholders is simplified to 3 actors: Onboard Controller or OC (which represents the vehicle), Fleet Operator or FO, and Provider of Services for UAM or PSU.

Table 1 Mapping Stakeholders from UML-4 Concept of Operations to UML-5 Concept of Operations

UML-4	UML-5
UAM Aircraft Automated Systems	Onboard Controller
UAM Aircraft	Onboard Controller
Pilot in Control (PIC)	Fleet Operator
Fleet Operator	Fleet Operator
PSU	PSU

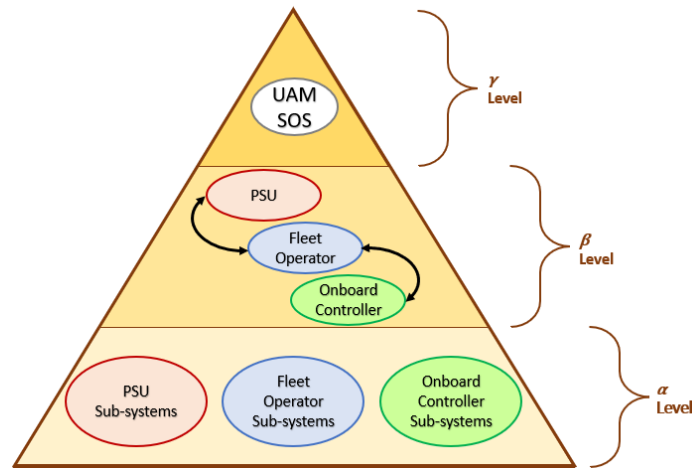


Fig. 1 Hierarchy of Systems

The levels in a ROPE table are indicated by Greek letters: α , β , γ . The α level contains subsystems of all the systems. The β level contains the systems themselves, i.e., Onboard Controller, Fleet Operator, and PSU. The γ level represents the entire UAM system-of-systems, encompassing the entire network of all the systems and their interactions. Our primary focus areas are the operations related to the β and γ levels (Fig. 1). The black bi-directional arrows in the β level represent ports where signals are communicated between the systems. Onboard Controller and Fleet Operator have a communication channel between them, and so do PSU and Fleet Operator. Any communication between Onboard Controller and PSU is assumed to happen via Fleet Operator. This architecture enables the construction of separate State Machine Diagrams for each actor. Furthermore, compartmentalization enables the development of algorithms that

contribute to Secure and Safe Assured Autonomy (S2A2) while making fault-detection and problem-solving easier. Based on this hierarchic architecture, Table 2 shows the ROPE table for the UAM SoS.

Table 2 Resources, Operations, Policies, Economics Table for UML-5

	Resources	Operations	Policies	Economics
α	Subsystems of: OC (e.g., sensing systems, communication systems), FO (e.g., remote pilot system, communication systems), PSU	Operations of subsystems (e.g., operating sensing systems, remote piloting, etc.)	Policies regarding the operations of these subsystems (e.g., standards for sensing and communication systems)	Subsystems' cost of operations, maintenance, acquisition, etc.
β	- Onboard Controller (OC) - Fleet Operator (FO) - Provider of Services for UAM (PSU)	- OC: Autonomous execution of vehicle operations such as state vector generation, sensing, maneuvering, etc. - FO: Receiving and evaluating passenger trip requests, acting as liaison between vehicle and PSU, and execution of fleet operations such as tracking and remote piloting. - PSU: Supporting Fleet Operators and maintaining UAM corridor safety through distribution of relevant data, verification of operational intent, etc.	- OC: Requirements specified for safe autonomy algorithms, secure autonomy algorithms, verification and validation algorithms. - FO: Regulatory compliance, policies for management, data sharing, situational awareness, and separation management of UAM operation. - PSU: Policies for maintaining safety, security, and efficiency of airspace, data sharing regulations, UAM corridor regulations, etc.	- OC: Vehicle operating costs - FO: fleet operations/maintenance costs; revenue from UAM trips. - PSU: Cost of PSU operations.
γ	UAM system of systems	Communication and cohesive operations of OC, FO, PSU	Policies regarding nominal UAM operations, and the mitigation of non-conformance and contingency cases	Economics of holistic UAM operations including the cost of operations, maintenance, and acquisitions, contracts between stakeholders, revenue generated through UAM services, etc.

IV. Block Definition Diagram (BDD) and Internal Block Diagram (IBD)

In the MBSE context, we use the ROPE table as a reference to create the SysML Block Definition Diagram (BDD). A BDD is used to define blocks in terms of their features and structural relationships with other blocks. The blocks represent the stakeholders in the ROPE table. Fig. 2 shows the primary stakeholders: Fleet Operator, PSU, Onboard Controller. Ports shown on the sides of the blocks are data planes and control planes (i.e., communication channels) between each stakeholder. For example, "PSU Port for FO" means the port is located in PSU block and communicates with FO block. Association arrows show how the Gamma level UAM system context block connects to Beta level blocks. Parameterized values are initialized under the Beta level blocks. These are user-defined and controlled values that determine how the sequence of activities under the stakeholders will execute. Some of the parameterized as enumeration literals (e.g., OCState, FOState, PSUState, OpIntentStatus). Also, Fig. 4 shows the BDD for a collection of signals that are communicated among the stakeholders. So, when a specific signal is communicated to an actor, the respective signal triggers an action in the activity diagrams. Fig. 3 shows the Internal Block Diagram (IBD) of the Gamma level UAM System Context. IBD contains an instance of the Beta level actors and answers the question of how the Beta level stakeholders are interconnected. The OC communicate to FO via a port, and vice-versa. FO communicates with PSU,

and vice-versa. But, there is no port between PSU and OC, and information to each other is hopped via FO.

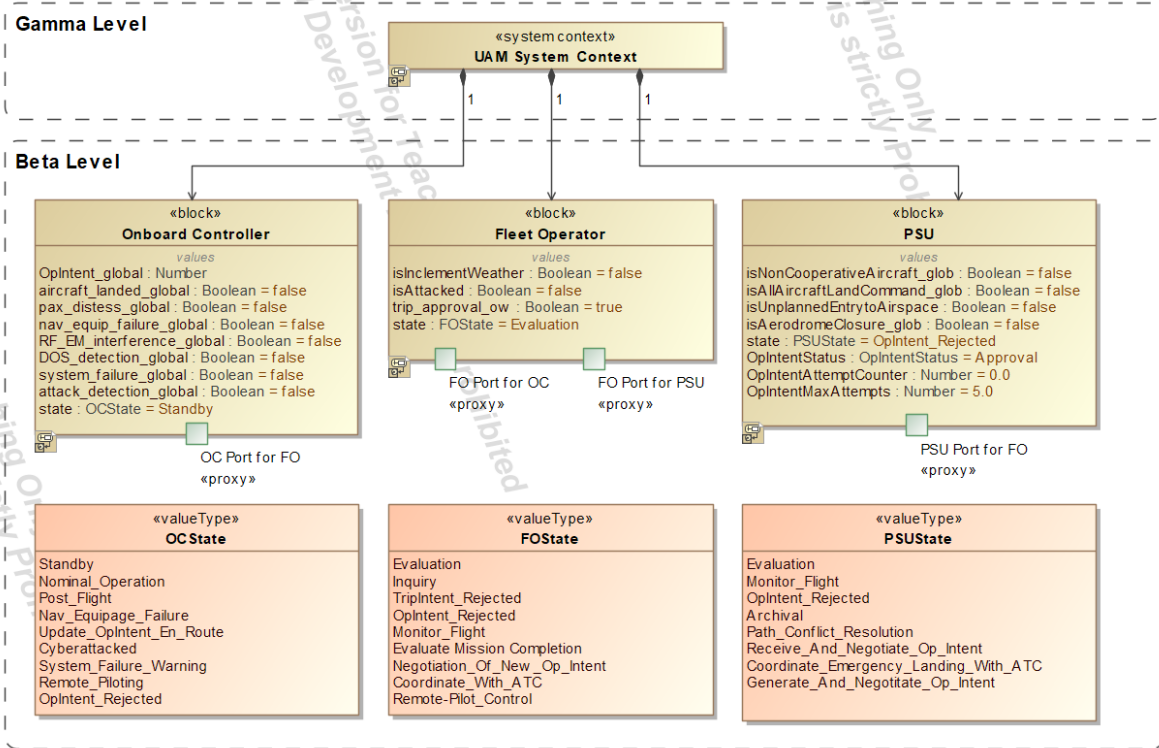


Fig. 2 Block Definition Diagram of UAM Stakeholders with Ports used for Communications

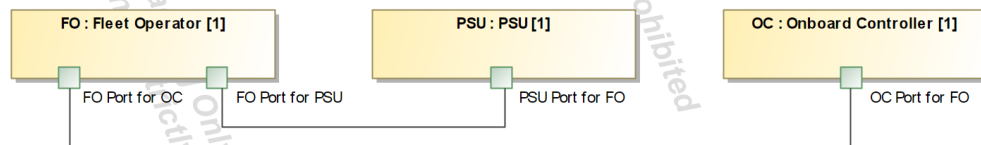


Fig. 3 Internal Block Diagram Showing Interconnections of UAM Stakeholders

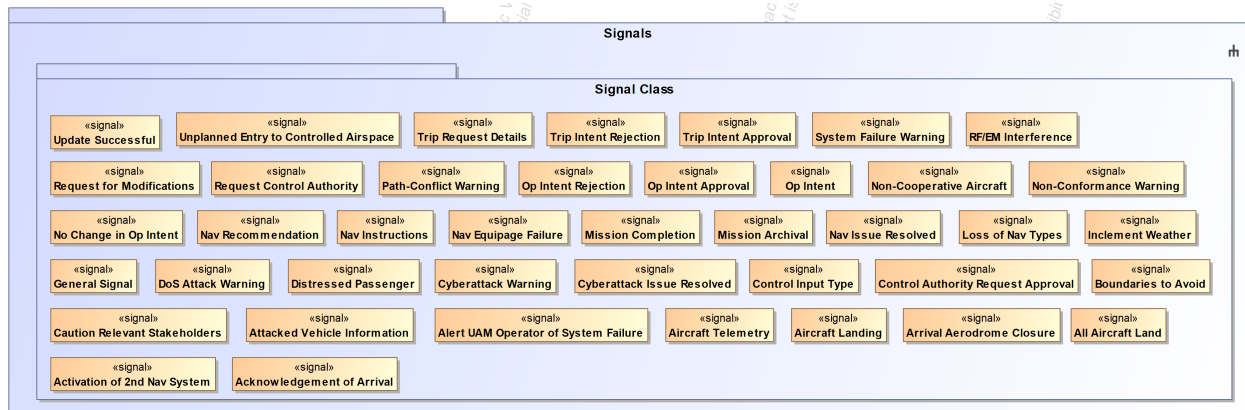


Fig. 4 Block Definition Diagram Signals used in Communications between UAM Stakeholders

V. State Machine Diagrams

A 'State Machine Diagram' can be used to describe the operating cycle of the actors in the UAM SoS. A 'state' is the condition or status of an actor at a given instance of time, which can be characterized by certain criteria or parameters. During its operation, an actor can switch states, i.e., undergo 'state transitions' which are initiated by specific triggers or events. A State Machine Diagram describes the state-based behavior (i.e., states and state transitions) of the actor throughout its operating cycle. The State Machine Diagrams for the UAM System-of-Systems Context identify the nominal and off-nominal states of the Onboard Controller, Fleet Operator, and PSU, based on the released Concept of Operations from NASA [2] and the FAA [1]. The UAM problem can be interpreted via different perspectives. This paper defines the operating cycle of each UAM actor and the overall SoS relative to a trip. Therefore, the high-level states of the State Machine Diagrams are *Pre-flight*, *In-flight*, and *Post-flight*. The non-conformance and contingency cases are considered separately, concurrent with the In-flight state.

The goal of developing State Machine Diagrams is to identify how the S2A2 algorithms for UML-5 fit into the UAM system-of-system context. This involves the identification of the operational context, such as use-case scenarios, and the state transition triggers that these algorithms must generate within this context. Figure 5 shows the State Machine Diagrams for the UAM system context (*gamma*-level). Figs. 6 to 8 present the diagrams for the *beta*-level actors, i.e., Onboard Controller, Fleet Operator, and PSU.

A. Gamma-Level State Machine Diagram

The γ -level State Machine Diagram (Fig. 5) encompasses the scope of the entire UAM system-of-systems, and thus it provides a high-level overview of the entire trip. The primary states are as follows:

- *Pre-flight*: Operations before the vehicle takes off (i.e., before acceptance of operational intent).
- *In-flight*: Operations while the vehicle is flying (i.e., after acceptance of operational intent).
- *Post-flight*: Operations after the vehicle has landed.

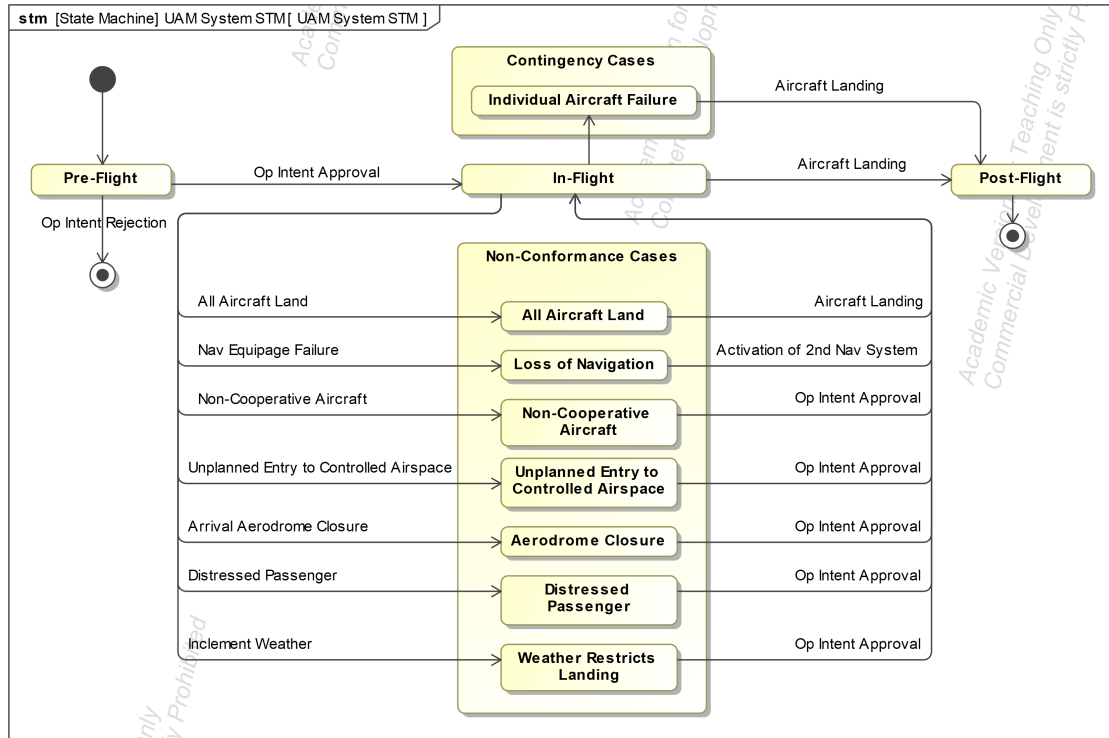


Fig. 5 Gamma Level State Machine Diagram of UML-5

The state transition from *Pre-flight* to *In-flight* occurs after the approval of operational intent (i.e., flight plan). If operational intent is rejected, the process is terminated. The *In-flight* state represents nominal flight. Once the aircraft lands, the state transition to *Post-flight* occurs. Besides these three states, there are two additional states to account for

the possibility of off-nominal conditions while the vehicle is flying. These are:

- *Non-Conformance Cases*: Off-nominal conditions that can be mitigated and reverted to a nominal state by taking an appropriate action. Specific types of non-conformance can be modeled as ‘sub-states’ or nested states under the main Non-Conformance Cases state. These states are triggered by predefined signals. Upon resolution of the problem, the system transitions to a nominal state. The non-conformance cases shown in Fig. 5 are:
 - *All Aircraft Land*
 - *Loss of Navigation*
 - *Non-Cooperative Aircraft*
 - *Unplanned Entry to Controlled Airspace*
 - *Aerodrome Closure*
 - *Distressed Passenger*
 - *Weather Restricts Landing*
- *Contingency Cases*: Off-nominal conditions are more severe and need special attention. Figure 5 only contains one contingency case for the γ level, i.e., *Individual Aircraft Failure*. This has been modeled as a sub-state of the *Contingency Cases* state. A System Failure Warning triggers the Contingency case, and in response, the aircraft must land as soon as possible to avoid a mishap.

The *All Aircraft Land* non-conformance case, and the *Individual Aircraft Failure* contingency case require immediate landing and therefore transition to the *Post-flight* state. All other off-nominal cases transition to the *In-flight* state after mitigation and the vehicle resumes nominal flight.

B. Beta Level State Machine Diagrams

The γ -level State Machine Diagram gives an overview of the SoS operating cycle, but the actual actions are executed by the β -level actors. These actors have been modeled as a ‘Block’ and they each have an associated variable named “state” which keeps track of the actor’s current state. Similar to the γ -level diagrams, the diagrams for each of the β -level actors consist of the following primary high-level states:

- *Pre-flight*
- *In-flight*
- *Post-flight*
- *Non-conformance Cases*
- *Contingency Cases*

However, many of these states have sub-states that are different for each actor. Additionally, the states (or sub-states) may be characterized by executable actions defined in an associated ‘Activity Diagram’. The State Machine Diagrams for the Onboard Controller, Fleet Operator, and PSU are discussed in the following sections.

1. Onboard Controller

Figure 6 shows The State Machine Diagram for Onboard Controller. The Onboard Controller is initially in a *Standby* state (nested under *Pre-flight*), during which it receives approval or rejection from the Fleet Operator in response to a trip request by a passenger. If PSU rejects the operational intent, then the Fleet Operator sends the corresponding rejection signal to the Onboard Controller, and the operation is terminated. If the PSU accepts the operational intent, then the Onboard Controller transitions to the *Nominal Operation* state, which is nested under the *In-flight* state. Regular activities such as sensing, telemetry, communication, state vector generation, etc., happen while the Onboard Controller is in this state. If no issues are encountered in the *In-flight* state, the aircraft lands, and the Onboard Controller transitions to the *Post-flight* state. However, if the Onboard Controller encounters any non-conformance or contingency situation, it transitions to the respective off-nominal state. The possible off-nominal conditions for the Onboard Controller are listed below.

- *Non-conformance*:
 - Navigational equipage failure or interference, causing loss of navigational capabilities.
 - Detection of an attack, that necessitates updating the operational intent en-route.
 - Cyberattack on the vehicle.
- *Contingency*: Failure of the aircraft systems, which requires the Onboard Controller to switch to the Remote Piloting state to help it land.

The triggers for all the state transitions depend on the value of the “state” variable. The S2A2 algorithms will generate the actions to be taken in case of each off-nominal situation.

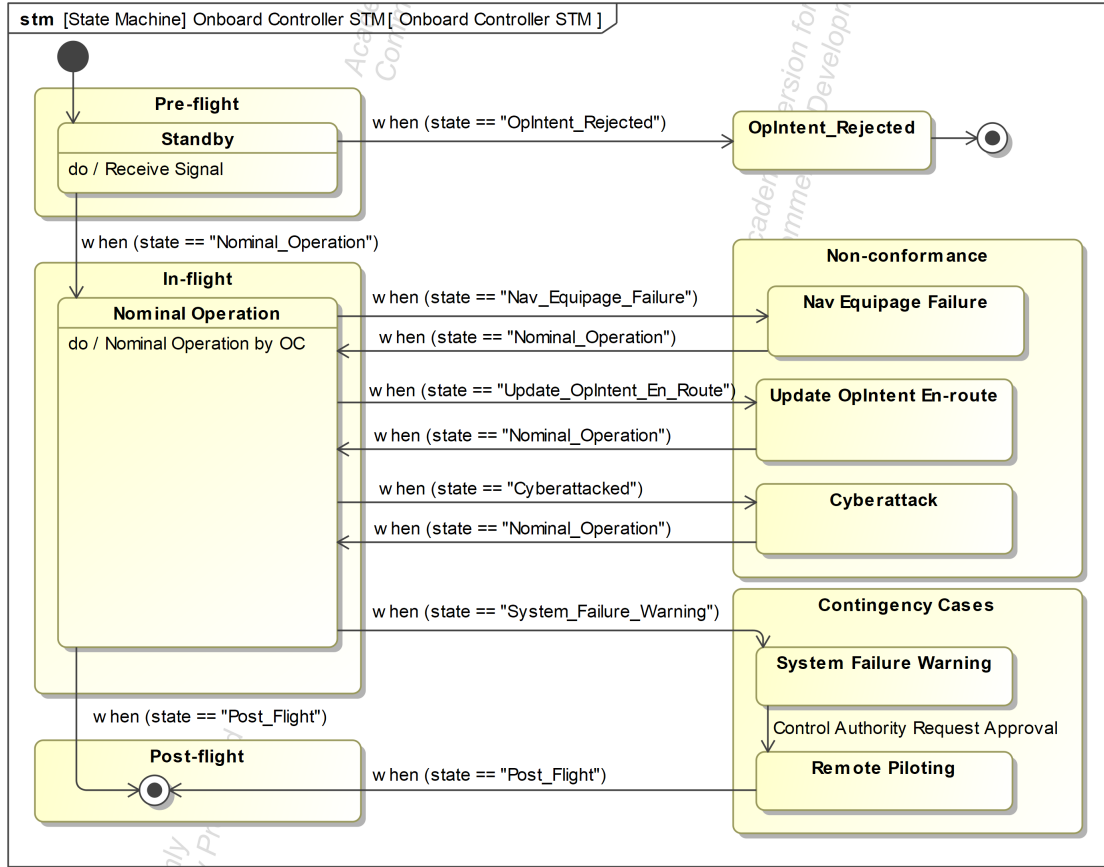


Fig. 6 Beta Level Onboard Controller State Machine Diagram of UML-5

2. Fleet Operator

The Fleet Operator State Machine Diagram is shown in Fig. 7. Here the *Pre-flight* state has two nested states, namely *Evaluation* and *Inquiry*. In the *Evaluation* state, the Fleet Operator receives a trip request from the passenger and processes it. If the request is rejected the operation ends. If it is accepted, then the Fleet Operator transitions to the *Inquiry* state and negotiates an operational intent (or flight plan) with the PSU. If the operational intent is rejected, then the process ends. If it accepted, the Fleet Operator transitions to the *Monitor Flight* state, nested under the *In-flight* state. While in this state, the Fleet Operator keeps the aircraft under observation and constantly checks for off-nominal conditions such as navigation loss, inclement weather, etc. In case of any such occurrence, the corresponding state transition is triggered to mitigate the problem. The off-nominal conditions for Fleet Operator are as follows.

- Non-conformance:
 - Negotiation of an new operational intent if a non-conformance warning is received from the Onboard Controller, if an attack or cyberattack is detected, if there is any path conflict or radio interference, or if the weather is inclement.
 - Coordinating with Air Traffic Control (ATC) in case of unplanned entry into controlled airspace.
- Contingency: Remote-Pilot Control if aircraft system failure is reported.

When the aircraft lands, the Fleet Operator transitions to the *Evaluate Mission Completion* sub-state of the Post-flight state. As with Onboard Controller, the state transitions for Fleet Operator too are triggered by the “state” variable.

3. PSU

The State Machine Diagram for PSU is shown in Fig. 8. While in the *Evaluation* state (nested under *Pre-flight*), the PSU evaluates the operational intent sent by the Fleet Operator. If the operational intent is rejected, the process ends. If it is accepted, then the PSU transitions to the *Monitor Flight* state nested under *In-flight*. In this state, the PSU

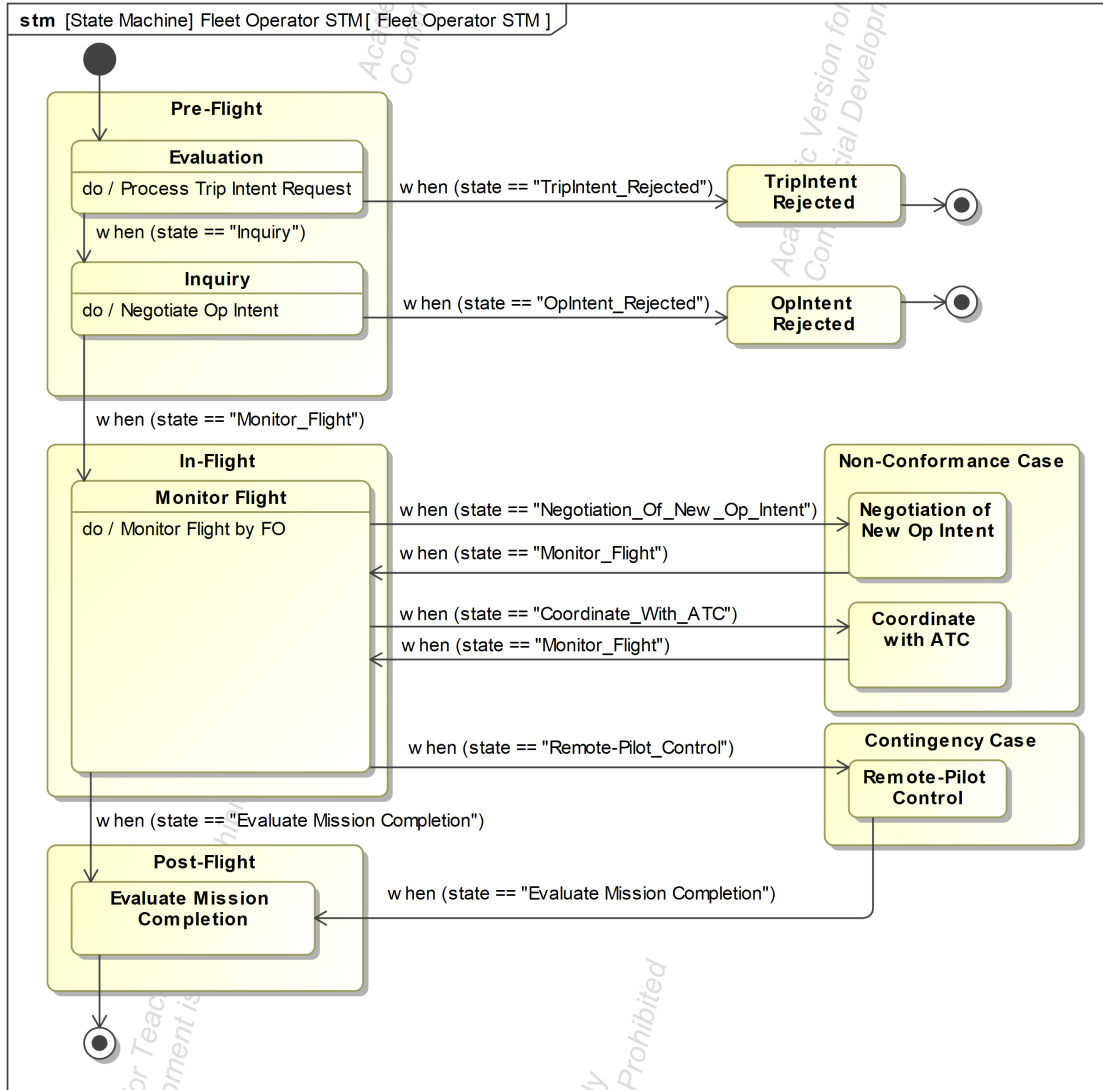


Fig. 7 Beta Level Fleet Operator State Machine Diagram of UML-5

oversees the UAM airspace and watches for off-nominal conditions. If any problem is detected, the PSU transitions to the corresponding non-conformance or contingency state. The off-nominal conditions for PSU are listed below.

- Non-conformance:
 - Receiving and Negotiating a new operational intent to mitigate unplanned entry into a controlled airspace.
 - Path-conflict resolution if an non-cooperative aircraft is detected.
 - Coordinating an emergency landing with Air Traffic Control if an “All Aircraft Land” notice is issued or there is any other type of non-conformance that makes flying risky.
- Contingency: Generation and negotiation of new operational intent for remote pilot control if an aircraft’s critical systems fail.

All the state transition are triggered by the “state” variable. After the aircraft lands, the PSU transitions to the *Archival* state under *Post-Flight* and the operating cycle terminates.

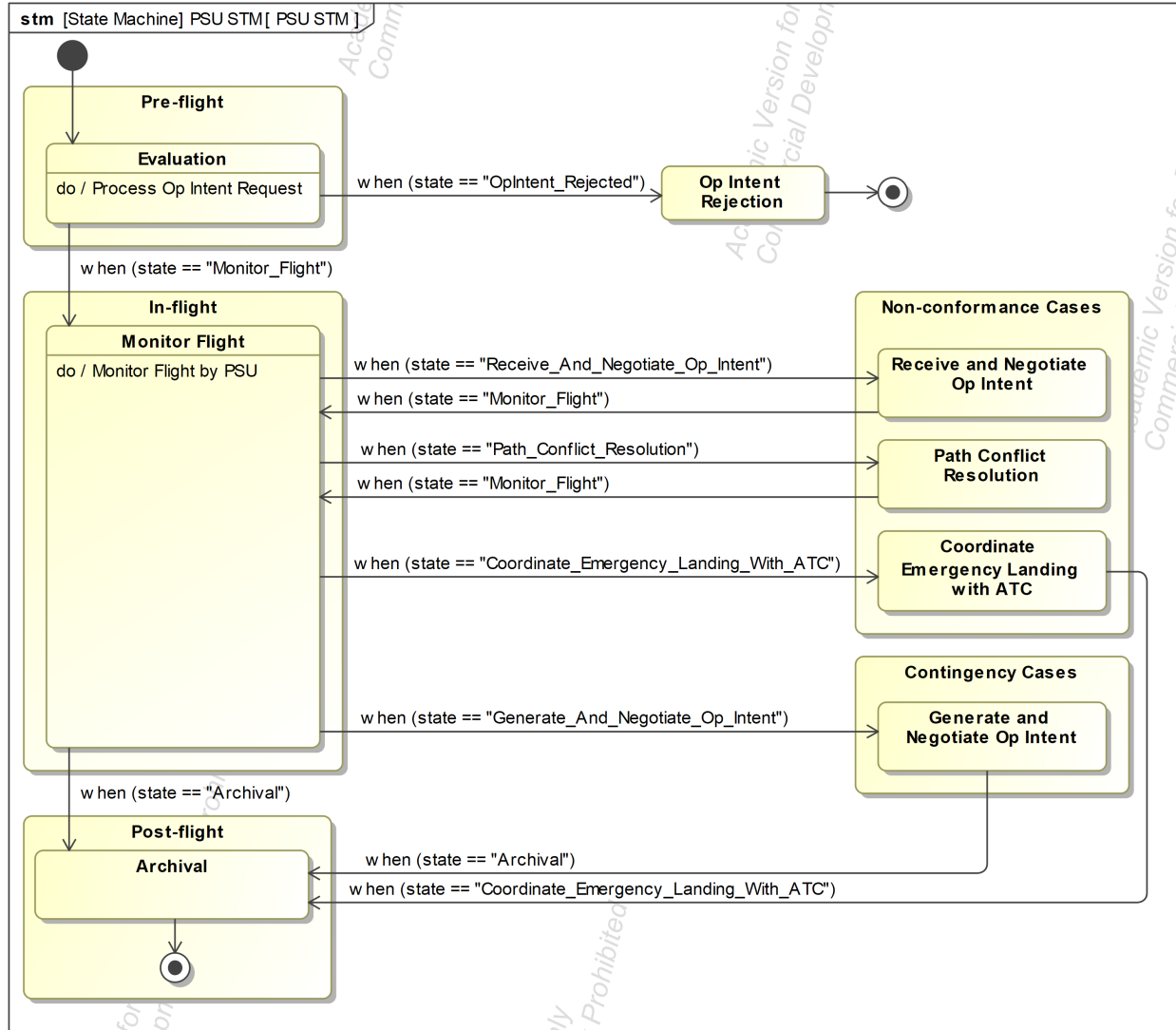


Fig. 8 Beta Level PSU State Machine Diagram of UML-5

VI. Onboard Controller Actions under Nominal Operations

The Nominal Operations of the Onboard Controller are described through two main Activity Diagrams:

- 1) Receive Signal: Associated with the *Standby* state nested within the Onboard Controller *Pre-flight* state.
- 2) Nominal Operation by OC: Associated with the *Nominal Operation* state nested within the Onboard Controller *In-flight* state.

A. Receive Signal

Fig. 9 shows the Activity Diagram of the Onboard Controller while the Passenger, Fleet Operator, and PSU actors are creating and deciding on the trip operation intent. During this period, the Onboard Controller is in a *Standby* state. If the Onboard Controller receives a signal from the Fleet Operator that the operational intent was accepted, the Onboard Controller transitions to the *Nominal Operation* state. If a signal of operational intent rejection is received from the Fleet Operator, the state transitions to *OpIntent Rejected*. The process ends after either of these transitions are made.

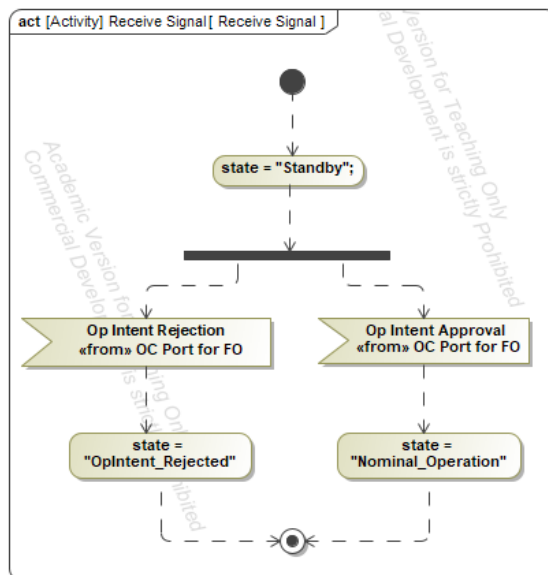


Fig. 9 ‘Receive Signal’ Activity Diagram

B. Nominal Operation by OC

Fig. 10 represents the Nominal Operation by OC Activity Diagram, which describes operations during flight. At this time the Onboard Controller is in the *Nominal Operations* state and actively notifies other actors of vehicle telemetry information before checking for various off-nominal events.

The process begins with the generation of a vehicle state vector that will be sent to other actors and used by the Onboard Controller to detect issues. Next the vehicle state vector to the Fleet Operator as an aircraft telemetry signal. The Onboard Controller then uses the V2V communication system to analyze neighboring aircraft telemetry and create a communication state vector. Raw sensing data relating to neighboring aircraft is then recorded from the onboard sensing systems. This information is used by the object detection algorithm to create a sensing state vector and obstacle state vector. Finally the data from the communication state vector and sensing state vector are packaged in the data fusion process and used by the Onboard Controller to check for off-nominal conditions. The series of off-nominal events, and the process taken if they are realized, are described sequentially below.

- 1) **System Failure Detection:** This process uses the vehicle state vector to detect whether the Onboard Controller system has failed. In the event a system failure is detected the Onboard Controller transitions to the *System Failure Warning* state, which is nested under the *Contingency Cases* state. From here the Onboard Controller will transition to the other nested state of *Remote Piloting* after receiving a Control Authority Request signal. The Onboard Controller remains here until the *Post-flight* state is triggered.
- 2) **DoS Attack Detection:** This process identifies if there has been a DoS attack on the communications channel using the vehicle, communications, and sensing state vectors. If a DoS attack is detected, an attack warning signal is sent to the Fleet Operator and the Onboard Controller transitions to the *Cyberattacked* state. This state is nested under the *Non-conformance* state and the Onboard Controller will remain there until the *Nominal Operations* state is triggered.
- 3) **Loss of Navigation Detection:** This process detects if there is a navigational equipment failure or radio frequency or electromagnetic interference. If either of these events are realized, the Onboard Controller transitions to the *Nav Equipment Failure* state, a nested *Non-conformance* state. The Onboard Controller then remains here until the *Nominal Operations* state is triggered.
- 4) **Attacked Vehicles Identification:** Using the vehicle, communication, and sensing system state vectors, this process identifies whether other vehicles have been attacked. In the event attacked vehicles are identified the Onboard Controller sends the information to the Fleet Operator and transitions into the *Update OpIntent En Route* state, which is a nested *Non-conformance* state where the Onboard Controller will remain until the *Nominal Operations* state is triggered.
- 5) **Dynamic and Static Obstacle Identification:** This process uses the safe motion planning and navigation

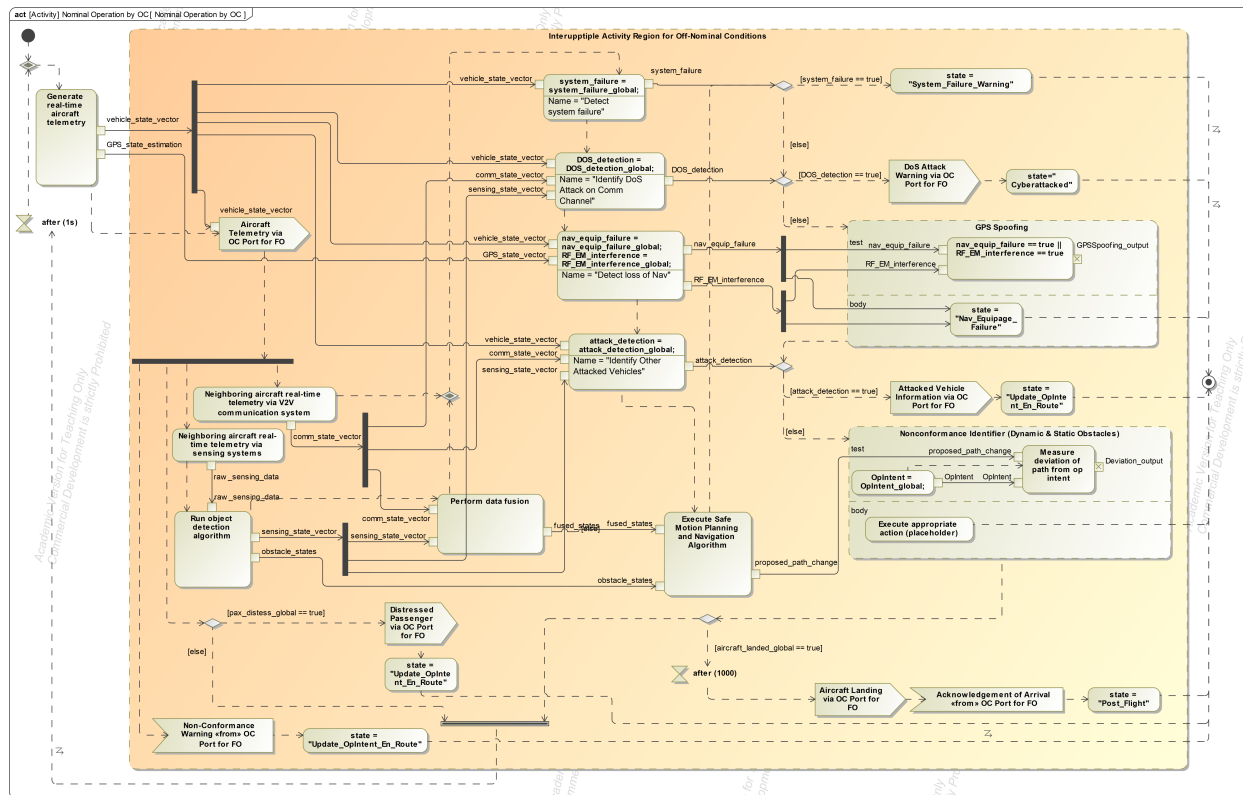


Fig. 10 ‘Nominal Operation by Onboard Controller’ Activity Diagram

algorithm to identify dynamic and static obstacles. This is related to the focus TC1-5, which falls under the jurisdiction of another university partner. As such, the diagram currently hosts a placeholder for executing the appropriate action according to the TC1-5 guidance. After this information has been obtained the Activity Diagram will be updated to fully display the process for this off-nominal condition.

- 6) **Passenger In Distress Detection:** This process occurs simultaneously to the above series of checks. If a passenger in distress is detected the Onboard Controller will transition to the *Update OpIntnt En Route* state.
- 7) **Non-Conformance Case Warning:** At any stage during the *Nominal Operation* state, if the Onboard Controller receives a non-conformance warning signal from the Fleet Operator it will transition to the *Update OpIntnt En Route* state.

After these checks have been completed, the final action of the diagram is to detect if the vehicle has landed. If this is realized as true the Onboard Controller notifies the Fleet Operator and waits to receive an acknowledgement of arrival before transitioning to the *Post Flight* state. The series of variable generations and off-nominal condition checks is repeated throughout the flight process. Once the Onboard Controller has transitioned to the *Post Flight* state, the State Machine Diagram terminates.

VII. Fleet Operator Actions under Nominal Operations

The activities performed by the the Fleet Operator during nominal operation of the aircraft are described using three Activity Diagrams:

- 1) Process Trip Intent Request: Associated with the *Evaluation* state nested under the *Pre-flight* state of Fleet Operator.
- 2) Negotiate Op Intent: Associated with the *Inquiry* state nested under the *Pre-flight* state of Fleet Operator.
- 3) Monitor Flight by FO: Associated with the *Monitor Flight* state nested under the *In-flight* state of Fleet Operator.

A. Process Trip Intent Request

This Activity Diagram, shown in Fig. 11, describes the required operations of the Fleet Operator when a passenger requests a trip. At this time, the Fleet Operator is in the *Evaluation* state. The Fleet Operator first receives a trip request, which includes details like the trip origin (passenger pickup point), destination, and desired time of departure. The trip is evaluated to see if it is feasible and if any aircraft in the fleet is available to take it. If the Fleet Operator decides that the trip can be accepted, then it transitions to the *Inquiry* state. Otherwise, it transitions to the *TripIntent Rejected* state and ends the operation.

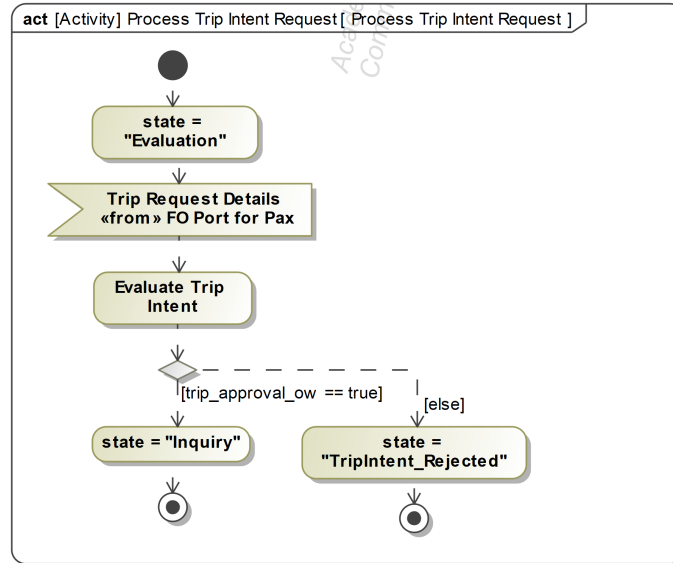


Fig. 11 ‘Process Trip Intent Request’ Activity Diagram

B. Negotiate Op Intent

This Activity Diagram, shown in Fig. 12, depicts the actions of the Fleet Operator when a trip request has been accepted. The Fleet Operator, which is now in the *Inquiry* state, generates an operational intent (i.e., a flight plan) which specifies waypoint markers (latitude, longitude, altitude, and time of passage) for the intended route. The operational intent is sent to the PSU for evaluation and the PSU’s response is received. If the operational intent is approved by the PSU, then Fleet Operator sends this information to the Onboard Controller and the passenger and the transitions to the *Monitor Flight* state. If the PSU does not find the operational intent feasible, then a modification request is sent to Fleet Operator. This forces the Fleet Operator to generate a new operational intent and send it to the PSU. If after a fixed maximum number of attempts the Fleet Operator is unable to generate a feasible operational intent then the request is rejected by PSU. In such a case, the Fleet Operator notifies the Onboard Controller and passenger and itself transitions to the *OpIntent Rejected* state to end the process.

C. Monitor Flight by FO

This Activity Diagram, shown in Fig. 13, describes the sequence of operations followed by the Fleet Operator when the vehicle is in flight. During this time, the Fleet Operator is in the *Monitor Flight* state, which implies that it observes the aircraft and tries to detect any off-nominal criteria. The possible off-nominal cases are summarized below, along with the standard activities to be performed if such cases are encountered.

- 1) **Activation of Second Navigation System:** The Fleet Operator receives this signal from the Onboard Controller and relays it to PSU.
- 2) **Inclement Weather:** If inclement weather is detected then the corresponding signals are sent to Onboard Controller and PSU. Fleet Operator then transitions to the *Negotiation of New Op Intent* state, nested under *Non-conformance Cases*.
- 3) **Cyberattack:** In case of a cyberattack, a ‘Non-conformance Warning’ signal is sent to Onboard Controller and

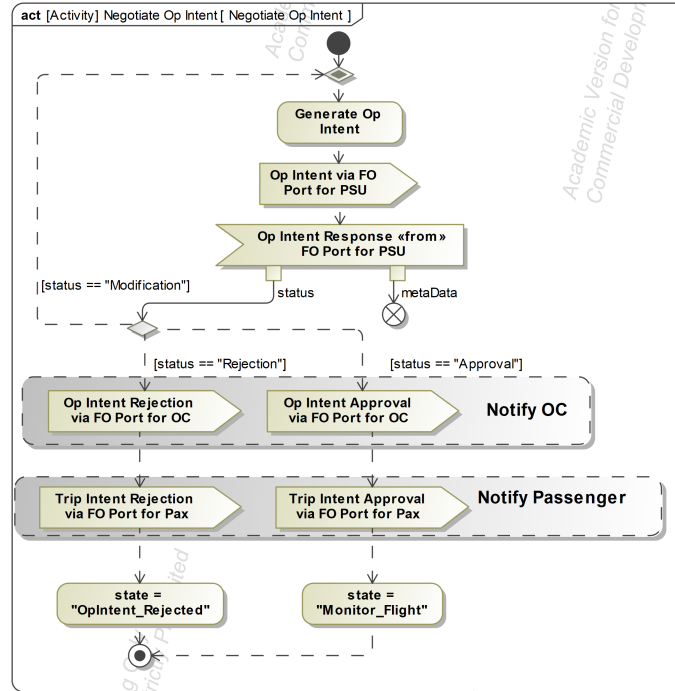


Fig. 12 ‘Negotiate Op Intent’ Activity Diagram

PSU, while the Fleet Operator transitions to the *Negotiation of New Op Intent* state.

- 4) **System Failure:** In case of any failure of the aircraft systems, the Fleet Operator requests Onboard Controller for control authority. Upon receiving it, a ‘System Failure Warning’ signal is sent to the PSU, while the Fleet Operator assumes control over the vehicle via remote pilot to attempt landing. Correspondingly, the state changes to *Remote-Pilot Control*, which is nested under *Contingency Case*.
- 5) **Unplanned Entry:** If the PSU notifies Fleet Operator of an unplanned entry into a controlled airspace, then this information is immediately relayed to the Onboard Controller. Meanwhile, Fleet Operator’s state changes to *Coordinate with ATC*.
- 6) **Vehicle Attacked:** If the Fleet Operator is notified of an attack by the Onboard Controller, it collects information on the attacked vehicle and passes it on to PSU. The state changes to *Negotiation of New Op Intent*, since the Fleet Operator and PSU must reformulate the flight plans.
- 7) **Non-conformance from OC:** If the Onboard Controller sends a generic ‘Non-conformance Warning’ signal to Fleet Operator, it signifies that the nature or source of has not yet been identified. In this case, the Fleet Operator notifies the PSU and changes its own state to *Negotiation of New Op Intent*.
- 8) **Path Conflict:** If the PSU informs the Fleet Operator of a path conflict between two vehicles, then Fleet Operator must notify the Onboard Controller and prepare to renegotiate the flight plan by changing its state to *Negotiation of New Op Intent*.
- 9) **RF/EM Interference:** If a radio-frequency or electromagnetic interference is encountered by an aircraft, the Onboard Controller sends a signal to Fleet Operator, which is then relayed to PSU. A mitigation attempt is made by changing the vehicle’s path. Therefore, the Fleet Operator’s state changes to *Negotiation of New Op Intent*.

It is to be noted that the Fleet Operator continuously monitors conditions like Second Navigation System Activation, Inclement Weather, and Cyberattack. This is why there are logic flow loops around those cases in Fig. 13. Whenever a state transition needs to be triggered, the “state” variable is assigned the appropriate value. If no adversity is faced during flight, the Fleet Operator remains in a nominal condition. When the Onboard Controller notifies the Fleet Operator of aircraft landing, the Fleet Operator acknowledges this and sends a signal to the PSU and the passenger. The state of the Fleet Operator then changes to *Evaluate Mission Completion*, nested under *Post-flight*.

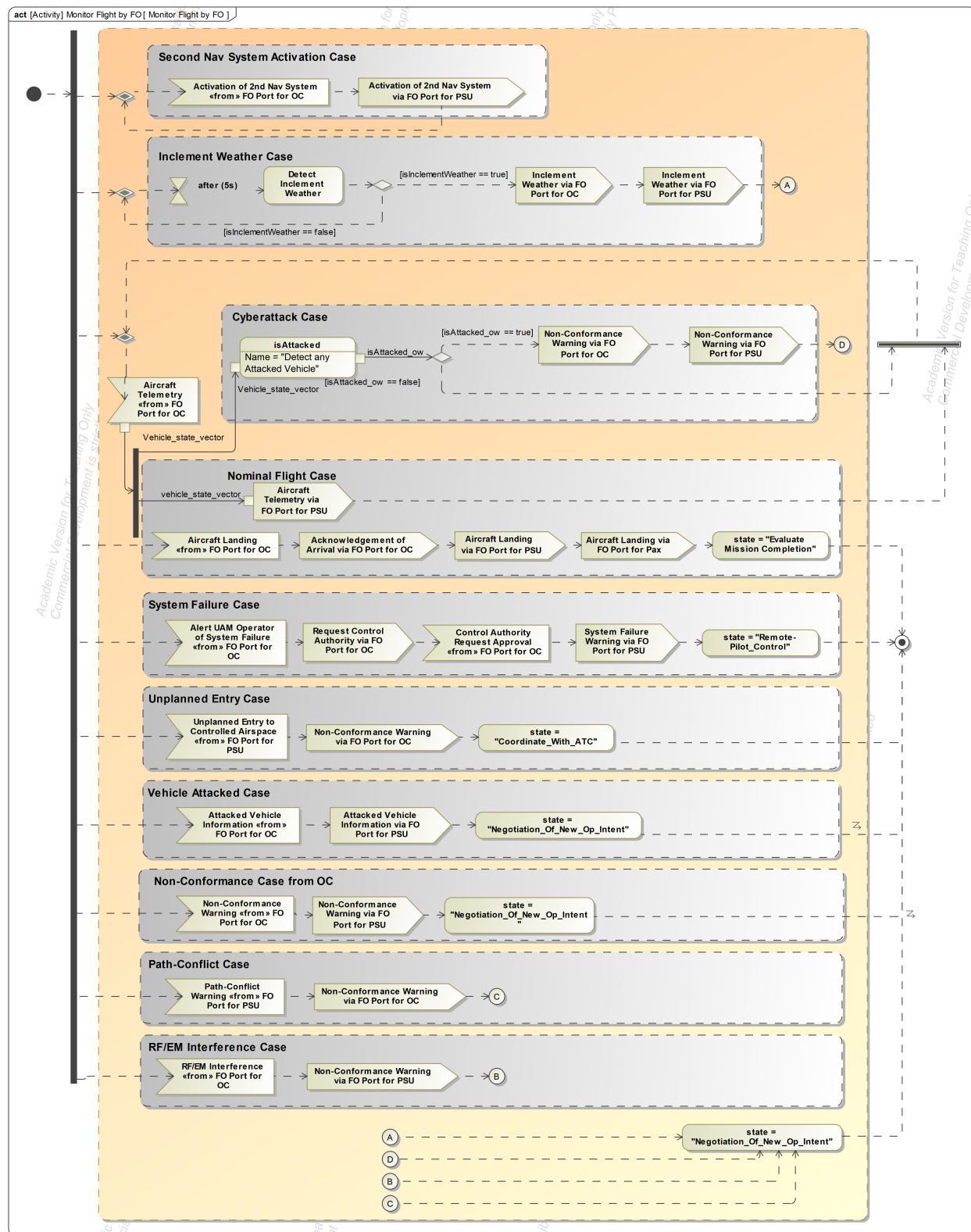


Fig. 13 'Monitor Flight by FO' Activity Diagram

VIII. PSU Actions under Nominal Operations

PSU State Machine Diagram is shown in Figure 8. Nominal operations for PSU includes Pre-flight, In-flight, Post-flight states. Each of these states includes an Activity Diagram. In summary, the Activity Diagrams are as follows:

- 1) Process Operational Intent Request
- 2) Monitor Flight by PSU

A. Process Operational Intent Request

Figure 14 shows the Activity Diagram of Process Op Intent Request inside the Evaluation state in pre-flight. The PSU receives operational intent from the Fleet Operator, and it begins evaluating it, considering various factors such as current UAM Corridor status, traffic density, weather, etc. The PSU allocates a limited number of resubmissions of the operational intent to avoid a Fleet Operator scheduling too much of the PSU's demanding slots. There are 3 outputs from this analysis:

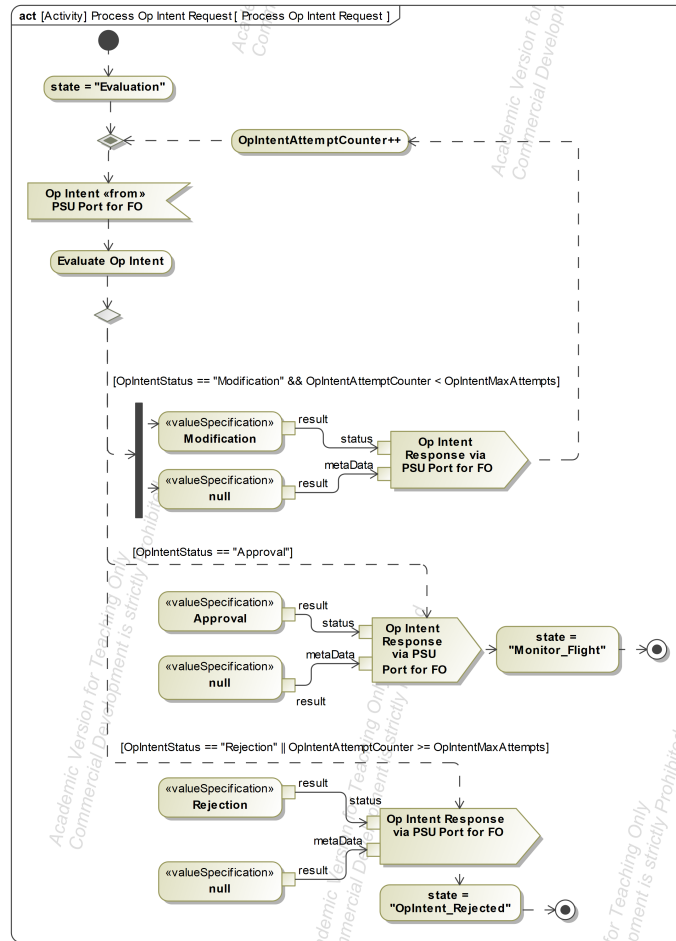


Fig. 14 'Process Op Intent Request' Activity Diagram

- 1) **OpIntentStatus = Approved:** PSU finds no conflicts of interest in the Fleet Operator submitted operational intent, and thus, approves it and notifies the Fleet Operator to proceed forward
- 2) **OpIntentStatus = Modifications:** PSU finds conflicts in the Fleet Operator submitted operational intent, and requests Fleet Operator for modifications and resubmission, as long as it is within Fleet Operator's limited number of attempts.
- 3) **OpIntentStatus = Rejection:** PSU finds conflicts in the Fleet Operator submitted operational intent, and since Fleet Operator exceeded its limited number of resubmissions, PSU rejects Op Intent, sends the signal to Fleet Operator regarding this

B. Monitor Flight by PSU

After Operational Intent is approved, PSU transitions from pre-flight and in-flight. PSU enters the “Monitor Flight by PSU” Activity Diagram. This Activity Diagram is shown in Figure 15. Once it receives Aircraft Telemetry from Fleet Operator, it loops through and checks for any non-conformance or contingency cases. If it detects any, it triggers the respective signal and initiates state transitions. Otherwise, it continues to loop through the Activity Diagram again and again. In addition to this, PSU also listens for any non-conformance or contingency signals from Fleet Operator or Onboard Controller. If PSU receives such signals, it initiates the respective non-conformance cases or enters post-flight case. The following are the non-conformance checks PSU does:

- 1) **All Aircraft Land Command from FAA:** if FAA says all aircraft must land, then PSU sends a non-conformance warning to Fleet Operator and to coordinate emergency landing with ATC
- 2) **Detect Non-cooperative Aircraft:** PSU sends to Fleet Operator regarding non-conformance warning
- 3) **Detect any unplanned entry into ATC airspace:** PSU sends to Fleet Operator regarding non-conformance warning, and PSU enters a state to resolve path conflict resolution
- 4) **Check for arrival aerodrome operation status:** if arrival aerodrome is closed, PSU alerts the Fleet Operator of the closure

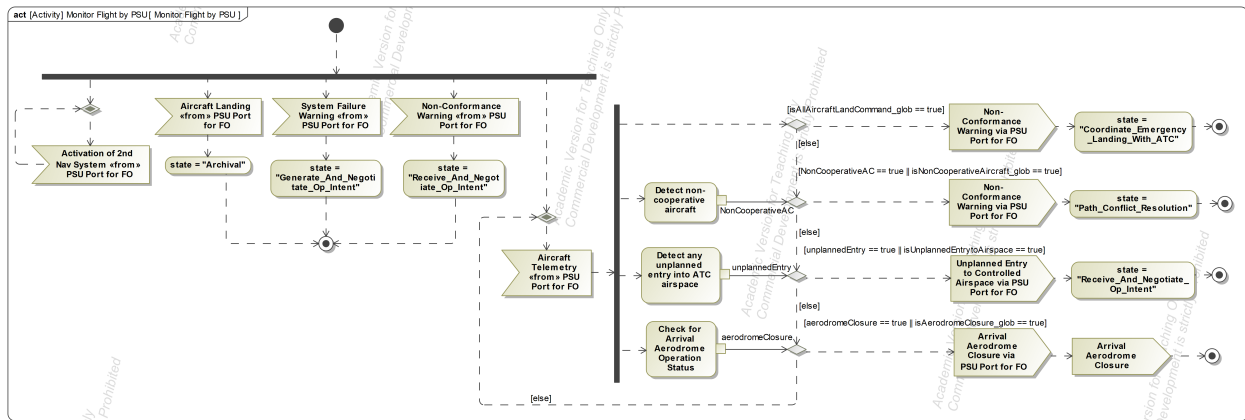


Fig. 15 ‘Monitor Flight by PSU’ Activity Diagram

The following are the non-conformance checks PSU receives from Fleet Operator or Onboard Controller. :

- 1) **System failure warning from FO:** once PSU receives system failure warning, it will try to generate and negotiate op intent
- 2) **Non-conformance warning from FO:** once PSU receives non-conformance warning, it will try to receive and negotiate op intent

IX. Conclusion & Future Work

This paper discusses the detailed activities that an onboard controller, a fleet operator, and a PSU must undertake for supporting and executing a nominal UAM flight. The State Machine Diagrams provide a quick overview of the stages that the actors can encounter under different nominal, non-conformance, or contingent scenarios. Furthermore, the Activity Diagrams show the detailed logical sequence and information flow between algorithms and actors. For example, when the fleet operator receives a trip request, the fleet operator needs to generate a flight intent and obtain approval from the PSU. This process involves algorithms to generate an operation intent and exchange operation intent between the PSU and the fleet operator.

The Activity Diagrams can serve as a quick drawing board for examining the integration of UAM-related algorithms. In the diagrams, the information flow can ensure that the interfaces between algorithms are consistent. The logical flows ensure that the algorithm execution sequence is correct. The Activity Diagrams ensure we are building the correct model.

The future work intends to integrate the MBSE model with other simulation software for assessing the fleet-level system safety level. Hence, the future work composes the following tasks.

- **Replace Activity Block with Real Algorithms:** The activity blocks in the Activity Diagrams represent what

algorithms need to accomplish. Hence, replacing the activity blocks with the actual algorithms for algorithm validation and verification is necessary.

- **Integration with Robotic Operating System:** The MBSE model is just a representation of the complex system-of-systems. Hence, the underlying algorithms still rely on some simulation environments. Since the Robotic Operating System (ROS) is one of the standard computational environments for the robotic algorithms, the team intends to integrate the ROS with the *MagicDraw* model
- **Integration with Third-Party UAM Simulator:** The ROS system can execute the control algorithms for enabling autonomous UAM operation. However, the control algorithms still need to receive the response from the physical model to determine the control commands. Hence, future work intends to integrate the third-party UAM simulator for modeling the response from the vehicles or the UAM traffic

The integrated tool can provide the capability of answering various safety-related research questions. A high-fidelity UAM simulator can ensure that the algorithms satisfy the minimum safety requirements. A lower fidelity and high-density UAM traffic simulator can verify if the algorithms can ensure a smooth operation between fleet operators, vertiports, and a PSU.

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